Structural comparison of Wild Type and Mutant

Comparison of CDS:

By comparing the CDS sequences of the WT PMM2 and the mutant PMM2 we have found that the mutation consists in a nucleotide substitution. In the 710th position of the CDS sequences the WT has a Thymine nucleotide, which is substituted by a Cytosine nucleotide in the mutant sequence (c.710T>C).

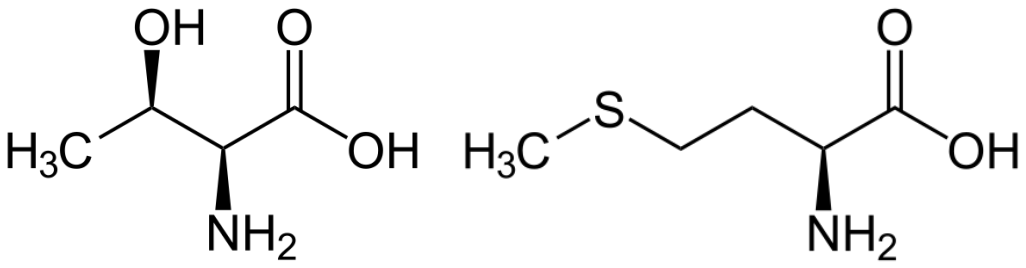
The affected nucleotide is the second element of the 237th codon (starting to count from the first start codon): ACG. The c.710T>C mutation causes it to become an ATG codon. The two codons, ACG and ATG, code for Threonine and Methionine respectively.

This means that the mutation is a missense one, as they code for different amino acids but the mutation doesn't cause a premature stop codon to appear.

Comparison of proteins:

Comparing the two protein sequences, we find that in the 237th position the WT protein has a Threonine amino acid, while the mutant has a Methionine acid (p.Thr237Met). This is consistent with the findings of the CDS comparison.

https://aminoacidsguide.com/Met.html



Threonine Methionine

Threonine is a polar molecule, with a hydrogen group as part of its side chain. Typically found in active sites, it can form hydrogen bonds and be phosphorylated.

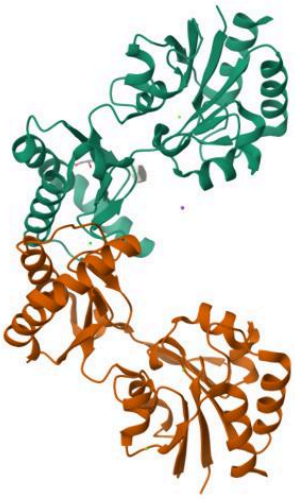
Instead, Methionine is a part of the aliphatic amino acid group. A nonpolar molecule with a sulfur atom that is part of a thioether group in its side chain. It cannot form hydrogen bonds, and it's hydrophobic.

The fact that they have different polarities could affect the interactions made by this position, the loss of the capacity to make hydrogen bonds could make the PMM2 less (or not at all) capable of its enzymatic function.

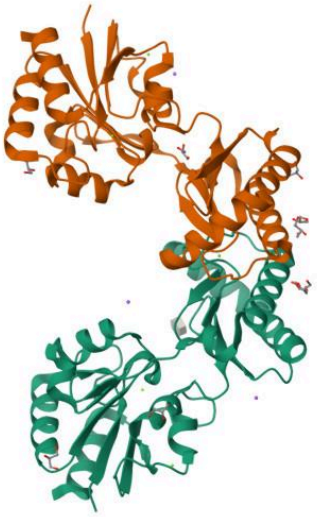
At the same time, the change in polarity and size could also affect protein folds. Threonine being hydrophilic while Methionine is hydrophobic means that they have opposite interactions with water, and the fact that Threonine is smaller could mean that Methionine doesn't fit and morphs the protein shape by trying to occupy the same space.

To further investigate how the mutation affects the proteins we compare their secondary structures.

Comparison of protein structure:



WT phosphomannomutase 2

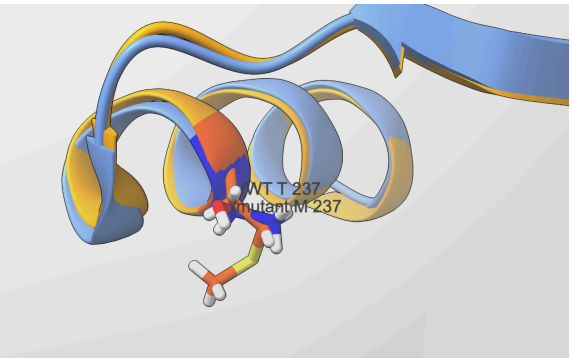


Mutated phosphomannomutase 2

Looking at their protein structure on PDB, they don't seem to be too different at a folding level, from which we deduce that the mutation has an effect at the chemical level of the protein, not the structural one.

PMM2 HTML file given by professors

Furthermore, reading the analysis done by the professors (cita?), we see that the local displacement at variant is of 0.262 Amperes, while the Human-ortholog local backbone displacement at the same aligned position is of 1.465 Amperes.



In the image on the left we see side to side the mutated and WT protein structures, with the alpha-helixes occupying almost the same space.

The main difference is in the specific amino acid occupying the 237th position.

The conclusion of this analysis is that the mutation is probably pathogenic due to its effects on the protein's interactions.